



NPWS

An tSeirbhís Páirceanna
Náisiúnta agus Fiadhúlra
National Parks and Wildlife Service

INSTALLING COIR ROLLS IN DEGRADED PEATLANDS

A STEP-BY-STEP FIELD INSTALLATION & QUALITY GUIDE



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1. Material Overview & Specifications

Coir rolls are adaptable, highly effective natural structures applicable to several types of degraded peatland areas. Manufactured from densely packed coconut husks wrapped securely in a biodegradable hessian (jute) net, they offer immediate physical stability while promoting long-term ecological recovery.

Specification	Standard Parameters
Material Composition	Coconut husks wrapped in a hessian (jute) net
Available Lengths	1.0 meter, 2.0 meters, and 3.0 meters
Standard Diameter	Approximately 30 cm
Primary Applications	Gully blocks in shallow gullies, baffles alongside peat hags, and stabilization in wide open bare peat areas.

Restoration Objectives & Benefits

Appropriate installation of coir rolls in targeted positions yields significant ecosystem dividends:

- **Sediment Control:** Measurable reductions in fine peat sediment loss.
- **Hydrology:** Increased localized water levels to encourage re-wetting.
- **Erosion Control:** Improved macro-soil stability against water and wind erosion.
- **Vegetation:** Natural vegetative regeneration across bare peat surfaces.

2. Installing Coir in Gully Channels

Coir logs are suited for installation in wide, shallow gullies where heavier traditional interventions like timber dams or stone blocks are inappropriate due to excessive gully width or high peat depth.

Log Length

Selecting the correct log length is important to prevent water from flowing around the log. The length of the coir log is dependent on the width of the gully:

$$\text{Required Coir Length} = \text{Gully Width} + \frac{1}{4} \text{ (25\%)}$$

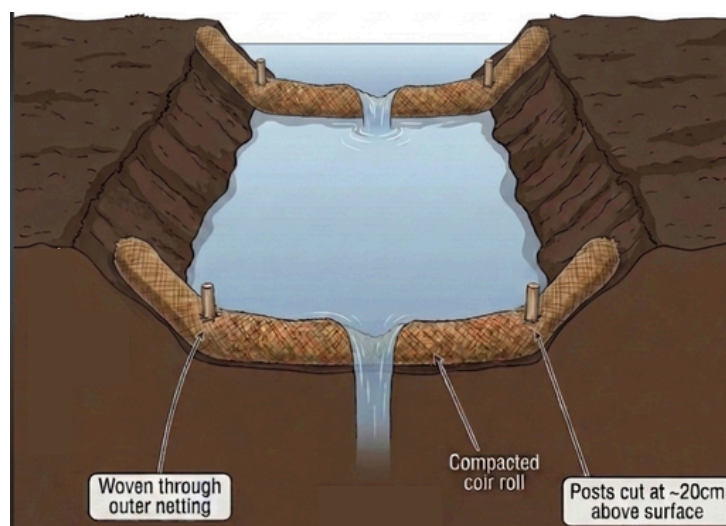
Note: The extra length is used to 'key' the coir log into the gully banks on both sides, minimising the risk of water flowing around log.

3. Field Installation Methods

3.1 Gully Channel Installation

Begin installation at the highest point of the gully where a block is deemed appropriate, and progress downstream.

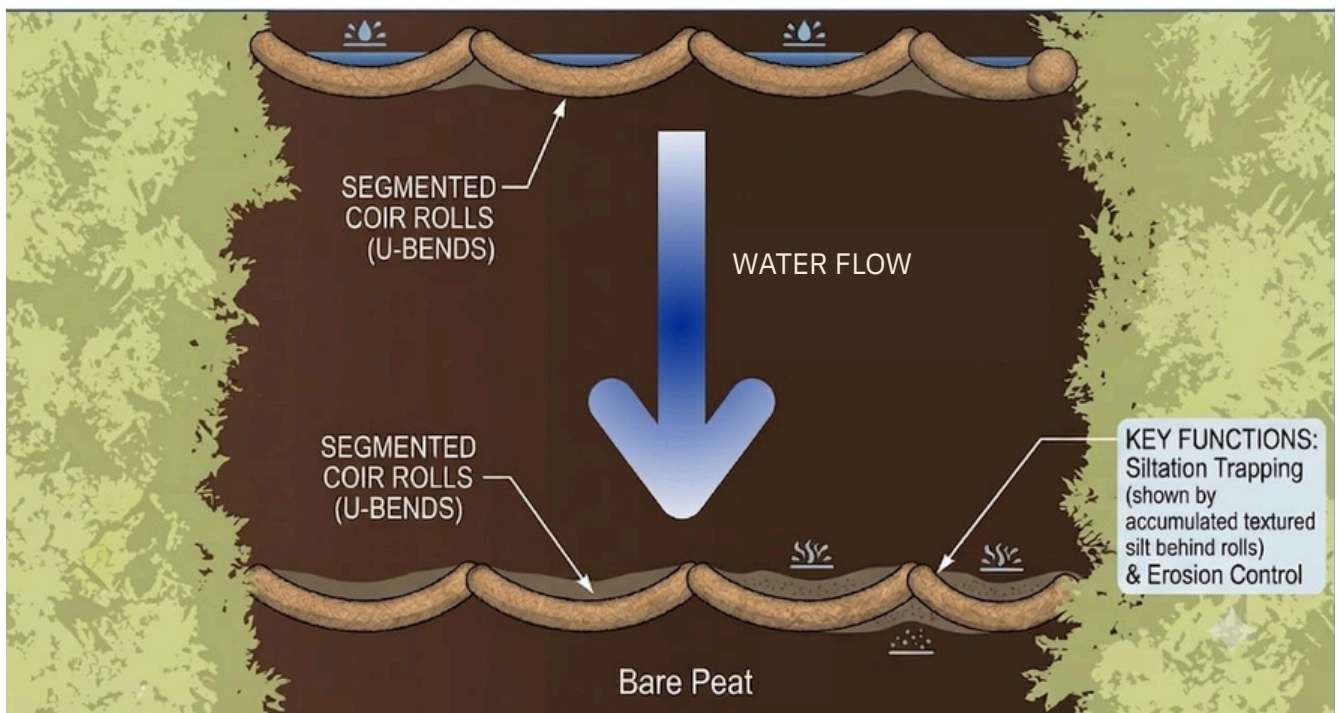
1. **Locate the Start Point:** Choose the highest elevation point in the gully where the channel depth is at least 15 cm.
2. **Excavate the Trench:** Dig a continuous trench across the gully bed. The trench should be approximately 10-15 cm deep, extend fully into the banks on both sides of the gully, and match the length of the coir log to be installed.
3. **Profile for Flow Control (U-Shape):**
 - **For 2m and 3m logs:** Excavate the center of the trench (the section at the center of the gully floor) slightly deeper. When the log is placed in the trench, it will then form a slight U-shape, ensuring water flows safely over the centre rather than scouring the banks.
 - **For 1m logs:** 1m logs are too short and stiff to bend into a U-shape, so the trench should maintain a uniform depth throughout.
4. **Drive Support Posts:** Place the coir log into the trench. Secure it firmly using 2-inch-diameter wooden posts driven vertically into the peat on the downstream side of the log. Weave the posts through 1 or 2 strands of the log's outer netting to anchor it.
 - **Post Allocation:** Use 2 posts for 1m and 2m logs; use 4 posts for 3m logs.
 - **Finishing Posts:** Cut the posts off at approximately 20 cm above the surface. A single standard 6-8ft timber post can be cut in half to supply two or more posts.
5. **Create the Upstream Seal:** Gather the excess peat excavated from the trench and compact it firmly on the upstream side of the coir roll. Press it hard against the log to form a tight, watertight seal between the coir roll and the gully bed.
6. **Progressive Sequence:** Install logs progressively down the length of the channel until the gully transitions into a profile better suited for timber dams (narrow and V-shaped) or stone dams (eroded down to bedrock).



3.2 Installation on Open Areas of Bare Peat (Scaling)

In cases where there are areas of bare peat with potential water flowing over, coir rolls can be placed in a scaling pattern across the peat to slow the water flow and trap sediment.

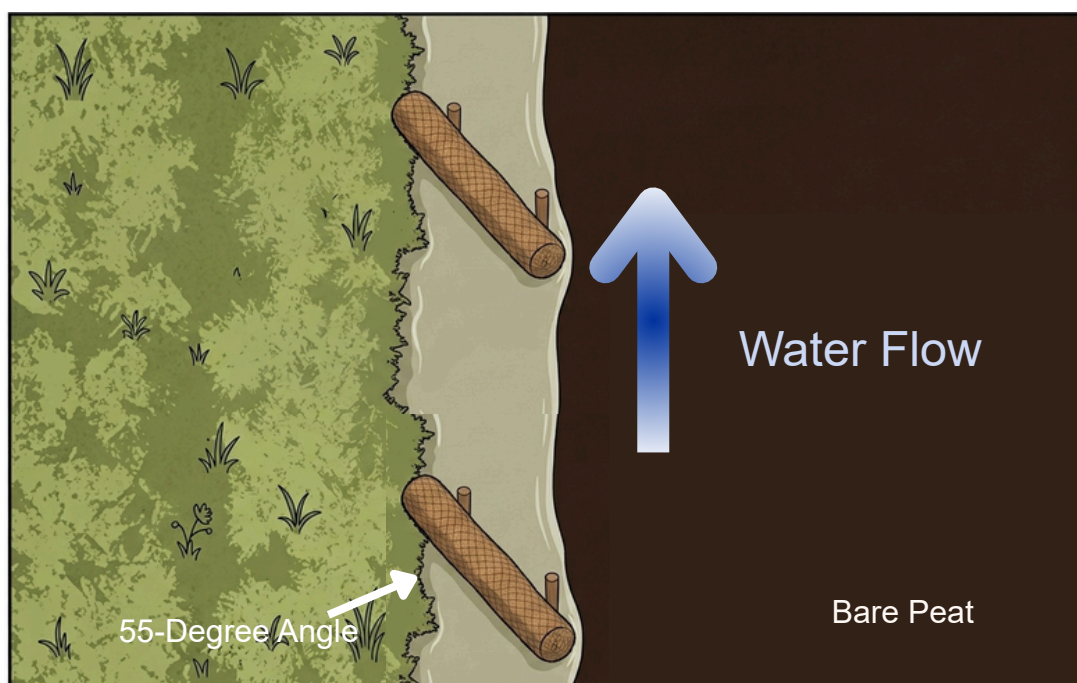
- **No Trenching Required:** For this installation approach, it is not necessary to 'bed' the rolls into the peat.
- **Orientation:** The rolls should be placed in rows across the bare peat area, perpendicular to the water flow.
- **Maximize Length:** Use the longest rolls available for the rows. For example, if the distance to be covered is 16m, use five 3m rolls and one 1m roll.
- **U-Shape Positioning:** Each roll should be placed in a slight horizontal U-shape, with the U positioned to trap flowing sediment.
- **Tight Joints:** In each row, butt the rolls together so that there is no water flow between the rolls.
- **Downstream Anchoring:** Wooden posts or pegs should be inserted on the downstream side of the rolls to secure the rolls in place. The posts/pegs should be woven into the mesh on the rolls.
- **Row Spacing:** The distance between each row is dependent on the slope and size of the bare peat area. A general rule of thumb is a 4-metre gap.



3.3 Installation as Baffles/Bunds

Coir can be used to create baffles and bunds that work to slow the flow of water. Unlike gully installations, where water is directed over the roll, water here is directed around the roll(s). This is useful at single peat banks (e.g., water flowing past peat hagsgs in an otherwise level area) to retain sediment and rebuild peat against the eroding bank.

- **Securing Mechanism:** As with the other installation methods, coir rolls are secured in place using wooden posts/pegs woven through the hessian mesh.
- **Roll Selection:** Shorter rolls are preferable in this situation—typically 1m or 2m rolls. Avoid 3m rolls, as they can become misshapen and dislodged during high water flows.
- **Angled Placement:** Rolls should be butted tight against the peat bank and installed at an approximate 50°–60° angle to the peat bank. This traps sediment effectively in the angle while slowing the water before it flows around the baffle.
- **Reinforcement:** As baffles are placed in areas where water flow can be high, extra securing of the rolls may be necessary.
- **Sequential Spacing:** Place baffles in sequence along the water flow path. Spacing depends on estimated water flow and slope angle, but a standard rule of thumb is 4 metres.



4. Quality Control & Field Checklist

Gully Blocks

- ☐ **Gully Topography Suitability:** Channel is wide and shallow, with a depth of at least 15cm at the initial upstream block.
- ☐ **Sizing Verification:** The chosen log length is equal to the gully width plus an additional one-quarter to allow proper bank seating.
- ☐ **Trench Execution:** Trench is 10-15cm deep, matches the full length of the log, and cuts deep into both side banks.
- ☐ **Crest Profile (U-Shape):** Centre of the trench is deepened for 2m and 3m logs to create a depression in the centre, directing overflow water and mitigating scouring of the gully banks.
- ☐ **Downstream Post Anchoring:** 2-inch diameter wooden posts are driven into the downstream side, woven through the structural netting strands, and trimmed to ~20cm above the log.
- ☐ **Post Quantities Checked:** At least 2 posts used for 1m or 2m logs, and 4 posts for 3m logs.
- ☐ **Upstream Compaction Seal:** Leftover peat is packed against the upstream face of the log to prevent water flowing under the structure.
- ☐ **Sequence Logic:** Installation begins at the highest suitable point and progresses systematically down the slope.

Open Peat & Baffles

- ☐ **Scaling Pattern Check:** Rows are arranged perpendicular to water flow, with the maximum log lengths used. These are tightly butted together, and curved into a U-shape against the flow.
- ☐ **Baffle Angle & Integrity:** Baffles are tight against the peat bank, aligned at a 50°–60° angle, with 1m/2m rolls used.
- ☐ **Spacing Standards:** Rows of scaling logs and sequential baffle installations follow a ~4-meter rule of thumb, unless the site topography necessitates different spacing.